

Machine Learning

(Assignment - 2)

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BITS Pilani - Work Integrated Learning Programmes

ML Classification Model Comparison

Breast Cancer Wisconsin (Diagnostic) Dataset

1. GitHub Repository Link

The complete source code, trained model files, test data, requirements.txt, and README.md are hosted in the following public GitHub repository:

GitHub Repository:

https://github.com/SunagP/ML_Assignment_2

Repository Contents:

- **app.py** - Streamlit web application with interactive dashboard
- **requirements.txt** - Python dependencies (streamlit, scikit-learn, numpy, pandas, matplotlib, seaborn)
- **README.md** - Complete project documentation with metrics, observations, and instructions
- **test_data.csv** - Held-out test data (114 samples, 31 columns)
- **model/train_models.py** - Training and evaluation script for all 5 classifiers
- **model/*.pkl** - Saved trained model files, scaler, feature names, and metrics
- **.gitignore** - Git ignore rules

2. Live Streamlit App Link

The application is deployed on Streamlit Community Cloud and is accessible via the following URL. It opens an interactive frontend when clicked:

Live Streamlit App:

<https://sunagp-ml-assignment-2.streamlit.app/>

App Features:

- CSV file upload for custom test data
- Model selection dropdown (5 classifiers)
- Evaluation metrics comparison table with highlighting
- Grouped bar chart and radar chart for visual comparison
- ROC curves for all models
- Confusion matrix heatmap per model
- Detailed classification report per model
- Dataset explorer with statistics and class distribution

3. Screenshots - BITS Virtual Lab Execution Proof

The following screenshots demonstrate the assignment execution on the BITS Virtual Lab environment.

Screenshot 1: Model Training on BITS Virtual Lab

```

train_models.py
def main():
    test_df["target"] = y_test.values
    test_csv_path = os.path.join(PROJECT_DIR, "test_data.csv")
    test_df.to_csv(test_csv_path, index=False)
    print(f"Test data saved -> {test_csv_path} ({len(test_df)} rows)")

    classifiers = {
        "Logistic Regression": LogisticRegression(max_iter=10000, random_state=42),
        "Decision Tree": DecisionTreeClassifier(random_state=42),
        "KNN": KNeighborsClassifier(n_neighbors=5),
        "Naive Bayes": GaussianNB(),
        "Random Forest": RandomForestClassifier(n_estimators=200, random_state=42),
    }

    trained_models = {}
    all_metrics = {}

    Dataset shape : (569, 31)
    Features      : 30
    Classes       : ['malignant', 'benign']

    Test data saved -> /home/cloud/downloads/ML Assignment 2-main/test_data.csv (114 rows)

    Logistic Regression
    Accuracy : 0.9825
    AUC      : 0.9954
    Precision: 0.9825
    Recall   : 0.9825
    F1       : 0.9825
    MCC      : 0.9823
  
```

Figure 1: Training all 5 ML models on BITS Virtual Lab (train_models.py execution)

Screenshot 2: Streamlit App Running on BITS Virtual Lab

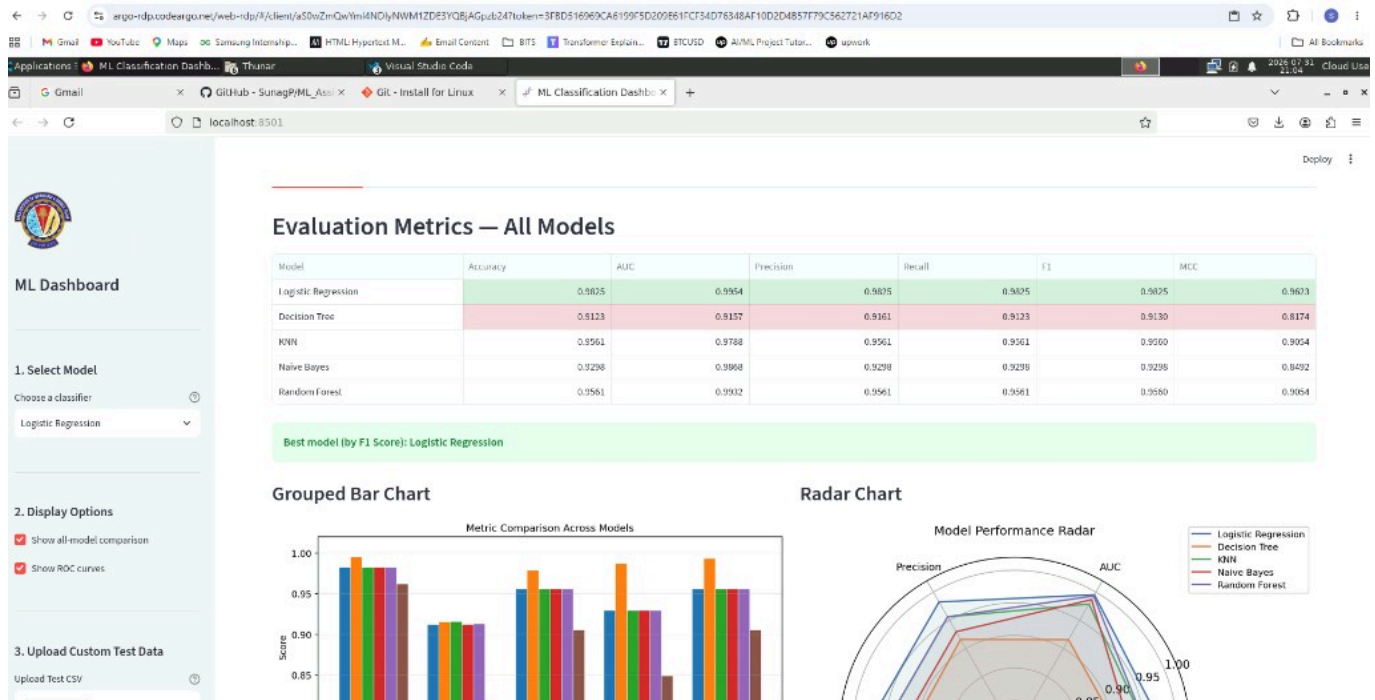


Figure 2: Streamlit ML Dashboard running on BITS Virtual Lab showing metrics comparison

Screenshot 3: Dependency Installation on BITS Virtual Lab

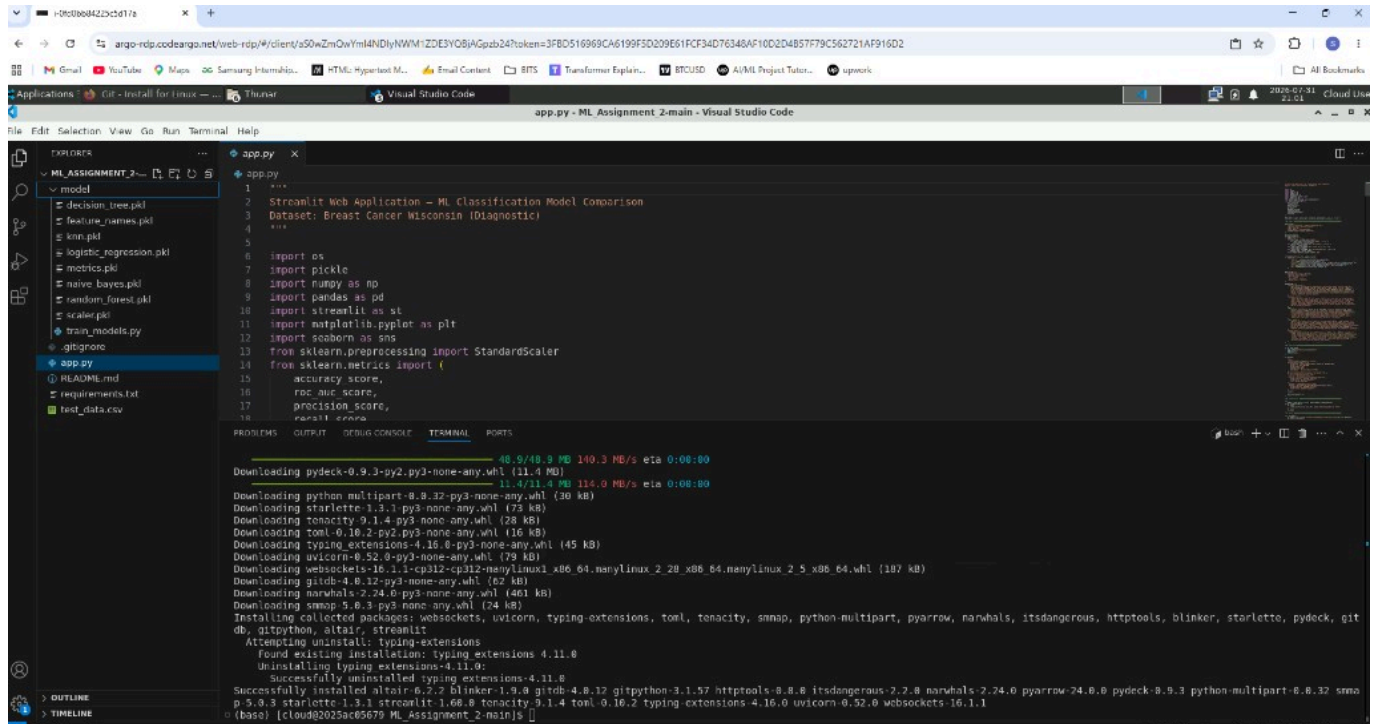


Figure 3: Installing project dependencies and setting up the environment on BITS Virtual Lab

4. GitHub README Content

a. Problem Statement

Breast cancer is one of the most prevalent cancers worldwide. Early and accurate diagnosis is critical for improving patient outcomes. The objective of this project is to build and compare five classical machine-learning classification models that predict whether a breast tumour is Malignant (0) or Benign (1) based on 30 real-valued diagnostic features extracted from digitised images of fine-needle aspirate (FNA) biopsies.

Each model is evaluated on the same held-out test set using six standard metrics - Accuracy, AUC, Precision, Recall, F1 Score, and Matthews Correlation Coefficient (MCC) - to determine which algorithm generalises best on this dataset. The results are presented through an interactive Streamlit web application deployed on Streamlit Community Cloud.

b. Dataset Description

Property	Value
Name	Breast Cancer Wisconsin (Diagnostic)
Source	UCI ML Repository / sklearn.datasets
Total Instances	569
Training Set	455 (80%)
Test Set	114 (20%, stratified)
Number of Features	30 (all real-valued)
Target Variable	Binary: 0=Malignant, 1=Benign
Class Distribution	212 Malignant (37.3%), 357 Benign (62.7%)
Missing Values	None

Ten real-valued measurements are computed for each cell nucleus in the FNA image: Radius, Texture, Perimeter, Area, Smoothness, Compactness, Concavity, Concave Points, Symmetry, and Fractal Dimension. For each measurement, three statistics are recorded (mean, standard error, and worst), yielding 30 features in total. All features are standardised using StandardScaler before model training.

Feature Details

#	Base Measurement	Features Generated
1	Radius	mean radius, radius error, worst radius
2	Texture	mean texture, texture error, worst texture
3	Perimeter	mean perimeter, perimeter error, worst perimeter
4	Area	mean area, area error, worst area
5	Smoothness	mean smoothness, smoothness error, worst smoothness
6	Compactness	mean compactness, compactness error, worst compactness
7	Concavity	mean concavity, concavity error, worst concavity
8	Concave Points	mean concave points, concave points error, worst concave points
9	Symmetry	mean symmetry, symmetry error, worst symmetry
10	Fractal Dimension	mean fractal dim, fractal dim error, worst fractal dim

c. GitHub Repository Link

Repository:

https://github.com/SunagP/ML_Assignment_2

Live Streamlit App:

<https://sunagp-ml-assignment-2.streamlit.app/>

d. Models Used

Evaluation Metrics Comparison Table:

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9825	0.9954	0.9825	0.9825	0.9825	0.9623
Decision Tree	0.9123	0.9157	0.9161	0.9123	0.9130	0.8174
KNN	0.9561	0.9788	0.9561	0.9561	0.9560	0.9054
Naive Bayes	0.9298	0.9868	0.9298	0.9298	0.9298	0.8492
Random Forest	0.9561	0.9932	0.9561	0.9561	0.9560	0.9054

Model Performance Observations

Logistic Regression

Best overall performer across every metric. Achieves the highest accuracy of 98.25%, the highest AUC of 0.9954, and the highest MCC of 0.9623. Its F1 score of 0.9825 is 5.27 percentage points higher than the next-best model (KNN/Random Forest at 0.9560). Only 2 out of 114 test samples were misclassified. The strong performance indicates that the 30 standardised features create a nearly linearly separable decision boundary, which Logistic Regression exploits optimally.

Decision Tree

Weakest performer among all five models. Accuracy is 91.23%, which is 7.02 percentage points lower than Logistic Regression. AUC is only 0.9157 - the lowest by a large margin (gap of 0.0797 vs Logistic Regression). MCC of 0.8174 is also the lowest, trailing Logistic Regression by 0.1449. The single unpruned tree overfits training data and produces axis-aligned splits. Misclassified 10 out of 114 test samples.

KNN

Achieves accuracy of 95.61% and MCC of 0.9054, placing it second alongside Random Forest. AUC of 0.9788 is solid but 0.0166 below Logistic Regression. Benefits from StandardScaler normalisation; with k=5 it captures local neighbourhood patterns. However, the 2.64 percentage-point accuracy gap vs Logistic Regression suggests the boundary is more global/linear. Misclassified 5 out of 114 test samples.

Naive Bayes

Gaussian Naive Bayes records accuracy of 92.98% and AUC of 0.9868. Its AUC is higher than KNN's (0.9788) and only 0.0086 behind Logistic Regression, indicating excellent probabilistic ranking. However, the 0.5 threshold leads to 8 misclassifications out of 114, pulling accuracy 5.27 points below Logistic Regression. MCC of 0.8492 reflects correlated features violating independence.

Random Forest

Matches KNN with 95.61% accuracy, 0.9560 F1, and 0.9054 MCC. AUC of 0.9932 is second-highest - only 0.0022 below Logistic Regression. Compared to single Decision Tree, improves accuracy by 4.38 percentage points and AUC by 0.0775, confirming bagging and feature randomisation reduce overfitting. Misclassified 5 out of 114 test samples, same as KNN.

Overall Winner

Logistic Regression is the clear winner. It ranks #1 in all six metrics: Accuracy (0.9825), AUC (0.9954), Precision (0.9825), Recall (0.9825), F1 (0.9825), MCC (0.9623). The 30 standardised features create a nearly linearly separable space, making a regularised linear model optimal. Runner-up by AUC: Random Forest (0.9932). Runners-up by accuracy: KNN and Random Forest (0.9561).

Project Structure

```
ML_Project/  
|-- app.py                (Streamlit web application)  
|-- requirements.txt      (Python dependencies)  
|-- README.md            (Project documentation)  
|-- test_data.csv        (Held-out test data, 114 samples)  
|-- .gitignore           (Git ignore rules)  
|-- model/  
    |-- train_models.py   (Training & evaluation script)  
    |-- logistic_regression.pkl  
    |-- decision_tree.pkl  
    |-- knn.pkl  
    |-- naive_bayes.pkl  
    |-- random_forest.pkl  
    |-- metrics.pkl  
    |-- scaler.pkl  
    |-- feature_names.pkl
```

Technologies Used

Technology	Purpose
Python 3.12	Programming language
scikit-learn	ML model training & evaluation
Streamlit	Interactive web application framework
pandas	Data manipulation & analysis
NumPy	Numerical computing
matplotlib	Static visualisations
seaborn	Statistical data visualisation